SHOKRULLAH SOROSH

PH.D. CANDIDATE

Structural Engineering · Earthquake Engineering · Structural Dynamics

CONTACT

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LOCATION
San Diego, California

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CORE SKILLS

Testing & Instrumentation

- Shake-table testing
- System- and subsystem-level testing
- Sensor and instrumentation planning and deployment
- Experimental structural and nonstructural design

Analysis & Design

- Finite element analysis (*ABAQUS, SAP2000, ETABS, ATENA, ANSYS, OpenSees*)
- Performance-based design
- Resilience-based design
- Nonstructural component performance (*Hilti Profis, SP3*)

Data & Monitoring

- Stochastic and deterministic system identification
- Structural health monitoring
- Displacement estimation from accelerations
- Algorithms and programming (*MATLAB and Python*)

EDUCATION

PhD Candidate in Structural Engineering
University of California San Diego
Sep. 2020 – Mar. 2027
Advisor: Dr. Tara Hutchinson · GPA 3.90/4.0

M.S. in Structural Engineering
Istanbul Technical University
Sep. 2018 – Jun. 2020
Advisor: Dr. Ali Sari · GPA 4.0/4.0
Thesis: Lateral load behavior of core rocking wall and coupled rocking wall systems

B.S. in Civil Engineering
Istanbul Technical University
Sep. 2014 – Jun. 2018
GPA 3.99/4.0 · **Valedictorian** (Top student, Class of 2018)

AWARDS

ASCE Norman Medal
2026

Graduate Student Fellowship, UC San Diego
2020

Valedictorian, Istanbul Technical University
2018

Civil and structural engineer (Ph.D. candidate, UC San Diego) with over six years of hands-on graduate-level experience in the planning, design, construction, instrumentation, and execution of large-scale engineering projects, including full-scale, multi-story shake-table tests of the tallest cold-formed steel and mass-timber buildings ever tested. Experienced in working with multidisciplinary teams of researchers, technicians, contractors, and industry partners to take complex infrastructure from concept through construction and operation, and in analyzing large datasets to quantify system performance and deliver clear, evidence-based recommendations.

Skilled in nonlinear finite element modeling (ABAQUS), system identification, and programming (MATLAB, Python) to develop efficient, standardized analysis and design workflows, and familiar with code- and standards-based design and performance-based design of buildings and nonstructural components. Contributor to three major testing programs: CFS10, NHERI TallWood, and Converging Design. Recipient of the 2026 ASCE Norman Medal.

> *Seeking full-time opportunities in structural engineering design and consulting, with particular interests in nonlinear structural analysis, seismic assessment, and forensic engineering.*

RESEARCH EXPERIENCE

Graduate Research Assistant
Sep. 2020 – Present
University of California San Diego — *La Jolla, CA*

CFS10 cfs10.ucsd.edu

- Core member of the team conducting earthquake testing of one of the tallest cold-formed steel buildings.
- Contributed to design, construction, instrumentation, and experimental testing phases.
- Performed system identification and structural health monitoring using advanced parametric/stochastic approaches.
- Characterized the dynamic response of the building and its nonstructural components (NSCs).

NHERI TallWood nheritallwood.mines.edu

- Key member of the team conducting shake-table testing of the tallest mass-timber building ever tested.
- Contributed to design, construction, instrumentation, and testing phases of the project.
- Led global displacement estimation using double integration of measured accelerations.
- Characterized dynamic response of structural and nonstructural components (NSCs).

Converging Design tallwoodinstitute.org

- Collaborated on shake-table testing of a 6-story building.
- Contributed to planning, instrumentation, and experimental testing phases.
- Evaluated dynamic response of structural and nonstructural components (NSCs).

Visiting Student Researcher
Jul. 2018 – Sep. 2018
Tokyo Institute of Technology — *Tokyo, Japan*

- Researched seismic performance of pinned and rocking RC walls under Dr. Anil Wijeyewickrema.

TECHNICAL EXPERIENCE

Technical Office Intern · *Omran Holding Group* — **Jul. 2017 – Aug. 2017 · Istanbul, Turkey**

Field Intern · *Omran Holding Group* — **Jul. 2016 – Aug. 2016 · Istanbul, Turkey**

TEACHING EXPERIENCE

Teaching Assistant
Fall 2023 – Spring 2026
University of California San Diego — *La Jolla, CA*

SE 9 **Algorithms and Programming for Structural Engineering** — Dr. F. Kuester · Fall 2023, Winter 2025, Winter 2026

SE 131B **Finite Element Analysis: Beam and Shell Models** — Dr. P. Krysl · Fall 2024

SE 215 **Cable Structures** — Dr. Q. Zhu · Spring 2026

MENTORSHIP

- Summer Training Academy for Research Success (STARS), Summer 2022
- ENLACE Program at UC San Diego, Summer 2023, 2026

AFFILIATIONS

- American Society of Civil Engineers (ASCE)
- Earthquake Engineering Research Institute (EERI)
- Structural Engineers Association of California (SEAOC)
  All 2021 – present

JOURNAL PUBLICATIONS

12 Rivera, D., **Sorosh, S.** et al., Towards Cold-Formed Steel Modular Construction in Tall Building Systems: Investigation within a Landmark 10-story Building Shake Table Test Program. (*under review in Structures*).

11 **Sorosh, S.** et al., Evolution of Dynamic Characteristics during Shake-Table Testing of a Full-Scale 10-Story Cold-Formed Steel Framed Building (*under review*, *Journal of Structural Engineering*, *ASCE*).

10 Hutchinson, T. C., Zhang, J., **Sorosh, S.**, … Schafer, B. W. (2026). Benchmarking Multi-Hazard (Earthquake and Fire) Resilience of a 10-story Cold-Formed Steel Framed Building – Part 1: Program Overview and Seismic Behavior. (*under review in Engineering Structures*).

09 Singh, A., **Sorosh, S.**, …, Hutchinson, T. C., Schafer, B. W. (2026). Benchmarking Multi-Hazard (Earthquake and Fire) Resilience of a 10-story Cold-Formed Steel-Framed Building. Part 2: Nonstructural Component Seismic Performance. (*under review in Engineering Structures*).

08 Kumar, U., Allen, R. M., Hutchinson, T. C., Zhang, J., **Sorosh, S.** (2026). Ubiquitous Smartphone Sensing of Urban Infrastructure across California for Seismic Resilience. (*in preparation for submission to Science Advances*).

07 **Sorosh, S.**, Hutchinson, T. C., Ryan, K. L., Smith, K. W., Kovac, A., Zabet, S., & Pei, S. (2025). Experimental characterization of a full-scale stair system detailed to achieve seismic resiliency. *Earthquake Engineering & Structural Dynamics*. doi.org/10.1002/eqe.4376

06 Ji, R., **Sorosh, S.**, Lo, E., Norton, T. J., Driscoll, J. W., Kuester, F., … & Hutchinson, T. C. (2025). Application framework and optimal features for UAV-based earthquake-induced structural displacement monitoring. *Algorithms*, 18(2), 66.

05 Hutchinson, T. C., **Sorosh, S.**, Bull, D., Kovac, A., Lizundia, B., Ryan, K., Sari, A., Smith, K., Wang, X., & Zhou, Y. (2025). Seismic behavior, observed performance, and design of stair systems: State-of-the-art. *Journal of Earthquake Engineering*. doi.org/10.1080/13632469.2025.2523783

04 Pei, S., Ryan, K. L., Berman, J. W., van de Lindt, J. W., Pryor, S., Huang, D., Wichman, S., Busch, A., Roser, W., Wynn, S. L., Ji, Y., Hutchinson, T., **Sorosh, S.**, Zimmerman, R. B., & Dolan, J. (2024). Shake-table testing of a full-scale 10-story resilient mass timber building. *Journal of Structural Engineering*, 150(12), 04024183. doi.org/10.1061/JSENDH.STENG-13752

03 Altunsu, E., Güneş, O., Öztürk, S., **Sorosh, S.**, Sarı, A., & Beeson, S. T. (2024). Investigating structural damage in Hatay Province after the Kahramanmaraş–Türkiye earthquake sequence. *Engineering Failure Analysis*, 157, 107857. doi.org/10.1016/j.engfailanal.2023.107857

02 **Sorosh, S.**, Hutchinson, T. C., Ryan, K. L., Smith, K., Belvin, R., & Black, C. (2024). High-fidelity finite element modeling of the seismic response of prefabricated steel stairs. *Earthquake Engineering & Structural Dynamics*, 53(8), 2491–2510. doi.org/10.1002/eqe.4117

01 **Sorosh, S.**, & Sari, A. (2022). Numerical investigation of the lateral load behavior of core and coupled rocking walls. *Revista de la Construcción*, 21(1), 36–52. doi.org/10.7764/rdlc.21.1.36

SELECT PEER-REVIEWED CONFERENCE PAPERS *(over 25 peer-reviewed conference papers)*

08 Singh, A., **Sorosh, S.**, Zhang, X., Zhang, J., Cyizere Rukundo, F., Eladly, M., Ji, R., Rivera, D., Padgett, L., Meacham, B. J., Gernay, T., Emberley, R. L., Schafer, B. W., & Hutchinson, T. C. (2026). Physical damage–measured response correlation for nonstructural components under multi-hazard (earthquake and fire) demands in the CFS10 building. Proceedings of Structures Congress 2025, Boston, MA.

07 **Sorosh, S.**, Zhang, J., Saifullah, M. K., Skolnik, D., & Hutchinson, T. C. (2026). System identification of a 10-story cold-formed steel building: Effects of sensor distribution and ambient conditions. Proceedings of IMAC-XLIV, Palm Springs, CA.

06 **Sorosh, S.**, Zhang, J., Lotfizadeh, K., Haddadi, H., Swensen, D., Branum, D., Kohler, M. D., Guy, R., Skolnik, D., Saifullah, M. K., Schafer, B. W., & Hutchinson, T. C. (2026). Implementation strategies and utility of a multi-sensor suite in a full-scale 10-story shake table test program. Proceedings of the 13th NCEE, Portland, OR.

05 **Sorosh, S.**, Zhang, J., Kovac, A., Smith, K., Schafer, B. W., & Hutchinson, T. C. (2026). Influence of stair system flexibility on building vibration period: Insights from full-scale shake table tests. Proceedings of ATC/SPONSE Sixth International Workshop on Seismic Performance of Non-Structural Elements, Japan.

04 Zhang, J., **Sorosh, S.**, Ji, R., Schafer, B. W., & Hutchinson, T. C. (2026). Full-scale 10-story cold-formed steel building shake table test: Protocol development and performance. Proceedings of NCEE 2026, Portland, OR.

03 **Sorosh, S.**, Zhang, J., Kovac, A., Smith, K., & Hutchinson, T. C. (2025). Design strategies and modeling verification for steel stairs with drift-compatible connections. Proceedings of SEAOC 2025 Convention, San Diego, CA.

02 **Sorosh, S.**, Hutchinson, T. C., Smith, K., & Kovac, A. (2024). Advancing resilient stair systems with friction-based drift-compatible connections. Proceedings of WCEE 2024, Milan, Italy.

01 **Sorosh, S.**, & Sarı, A. (2019). Comparing moment capacities of damaged and undamaged tubular members. In IEEE ISMSIT (pp. 1–4). doi.org/10.1109/ISMSIT.2019.8932850